

In [1]:

```
pip install pandas
```

Requirement already satisfied: pandas in /opt/anaconda3/lib/python3.14/site-packages (3.0.3)

Requirement already satisfied: numpy>=2.3.3 in /opt/anaconda3/lib/python3.14/site-packages (from pandas) (2.4.6)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/lib/python3.14/site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in /opt/anaconda3/lib/python3.14/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Note: you may need to restart the kernel to use updated packages.

In [2]:

```
import pandas as pd
data=pd.read_csv(r"/Users/udaykumarreddy/Downloads/Mall_Customers.csv")
```

In [3]:

```
data
```

Out[3]:

|     | CustomerID | Genre  | Age | Annual Income (k\$) | Spending Score (1-100) |
|-----|------------|--------|-----|---------------------|------------------------|
| 0   | 1          | Male   | 19  | 15                  | 39                     |
| 1   | 2          | Male   | 21  | 15                  | 81                     |
| 2   | 3          | Female | 20  | 16                  | 6                      |
| 3   | 4          | Female | 23  | 16                  | 77                     |
| 4   | 5          | Female | 31  | 17                  | 40                     |
| ... | ...        | ...    | ... | ...                 | ...                    |
| 195 | 196        | Female | 35  | 120                 | 79                     |
| 196 | 197        | Female | 45  | 126                 | 28                     |
| 197 | 198        | Male   | 32  | 126                 | 74                     |
| 198 | 199        | Male   | 32  | 137                 | 18                     |
| 199 | 200        | Male   | 30  | 137                 | 83                     |

200 rows × 5 columns

In [4]:

```
data.head()
```

Out[4]:

|   | CustomerID | Genre  | Age | Annual Income (k\$) | Spending Score (1-100) |
|---|------------|--------|-----|---------------------|------------------------|
| 0 | 1          | Male   | 19  | 15                  | 39                     |
| 1 | 2          | Male   | 21  | 15                  | 81                     |
| 2 | 3          | Female | 20  | 16                  | 6                      |
| 3 | 4          | Female | 23  | 16                  | 77                     |
| 4 | 5          | Female | 31  | 17                  | 40                     |

In [5]:

```
data.tail()
```

Out[5]:

|     | CustomerID | Genre  | Age | Annual Income (k\$) | Spending Score (1-100) |
|-----|------------|--------|-----|---------------------|------------------------|
| 195 | 196        | Female | 35  | 120                 | 79                     |
| 196 | 197        | Female | 45  | 126                 | 28                     |
| 197 | 198        | Male   | 32  | 126                 | 74                     |
| 198 | 199        | Male   | 32  | 137                 | 18                     |
| 199 | 200        | Male   | 30  | 137                 | 83                     |

In [6]:

```
data.shape
```

Out[6]:

(200, 5)

In [7]:

```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Genre                  200 non-null   str
2   Age                    200 non-null   int64
3   Annual Income (k$)     200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), str(1)
memory usage: 8.9 KB
```

In [8]:

```
data.describe()
```

Out[8]:

|       | CustomerID | Age        | Annual Income (k\$) | Spending Score (1-100) |
|-------|------------|------------|---------------------|------------------------|
| count | 200.000000 | 200.000000 | 200.000000          | 200.000000             |
| mean  | 100.500000 | 38.850000  | 60.560000           | 50.200000              |
| std   | 57.879185  | 13.969007  | 26.264721           | 25.823522              |

|            | CustomerID | Age       | Annual Income (k\$) | Spending Score (1-100) |
|------------|------------|-----------|---------------------|------------------------|
| <b>min</b> | 1.000000   | 18.000000 | 15.000000           | 1.000000               |
| <b>25%</b> | 50.750000  | 28.750000 | 41.500000           | 34.750000              |
| <b>50%</b> | 100.500000 | 36.000000 | 61.500000           | 50.000000              |
| <b>75%</b> | 150.250000 | 49.000000 | 78.000000           | 73.000000              |
| <b>max</b> | 200.000000 | 70.000000 | 137.000000          | 99.000000              |

In [9]:

```
data.columns
```

Out[9]:

```
Index(['CustomerID', 'Genre', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='str')
```

In [10]:

```
data.dtypes
```

Out[10]:

```
CustomerID      int64
Genre           str
Age             int64
Annual Income (k$)  int64
Spending Score (1-100)  int64
dtype: object
```

In [11]:

```
data.isnull().sum()
```

Out[11]:

```
CustomerID      0
Genre           0
Age             0
Annual Income (k$)  0
Spending Score (1-100)  0
dtype: int64
```

In [12]:

```
data.duplicated().sum()
```

Out[12]:

```
np.int64(0)
```

In [13]:

```
data.nunique()
```

Out[13]:

```
CustomerID      200
Genre           2
Age             51
Annual Income (k$)  64
Spending Score (1-100)  84
dtype: int64
```

In [ ]: